A1
PICO 2 SC1632 SOCKETED
DOCK_TX 1
DOCK_RX 2
MCU_MOTOR_A_N 4
MCU_DS_B_N 5
MCU_DS_A_N 6
MCU_MOTOR_B_N 7
MCU_DIR_N 9
MCU_STEP_N 10
MCU_WDATA_N 11
MCU_WGATE_N 12
MCU_HEAD_N 14
MCU_DENSEL 15
ASSERT_INDEX 16
ASSERT_TRK0 17
ASSERT_WP 19
ASSERT_RDATA 20
ASSERT_CHANGE 21
SPI_CS_N 23
SPI_SCK 24
SPI_MOSI 25
SPI_MISO 26
MEDIA_ARM 27
INTERLOCK_OK 28
MEDIA_ERROR_N 31
LED_DATA 32
DOCK_BUSY 34
GND 3
GND 8
GND 13
GND 18
GND 23
GND 30
GND 35
GND 38
GND 39
3V3 36
VSYS 40
J1
34-PIN PC FLOPPY
GND 1
GND 3
GND 5
GND 7
GND 9
GND 11
GND 13
GND 15
GND 17
GND 19
GND 21
GND 23
GND 25
GND 27
GND 29
GND 31
GND 33
FDD_DENSEL 2
FDD_INDEX_N 4
FDD_MOTOR_A_N 6
FDD_DS_B_N 8
FDD_DS_A_N 10
FDD_MOTOR_B_N 12
FDD_DIR_N 14
FDD_STEP_N 16
FDD_WDATA_N 18
FDD_WGATE_N 20
FDD_TRK0_N 22
FDD_WP_N 24
FDD_RDATA_N 26
FDD_HEAD_N 28
FDD_CHANGE_N 34
U1
SN74LVC245AN / 5V→3V3 INPUT
3V3 1
GND 10
3V3 20
GND 19
FDD_MOTOR_A_N 2
MCU_MOTOR_A_N 3
FDD_DS_B_N 4
MCU_DS_B_N 17
FDD_DS_A_N 5
MCU_DS_A_N 16
FDD_MOTOR_B_N 6
FDD_DIR_N 7
FDD_STEP_N 8
FDD_WDATA_N 9
FDD_WGATE_N 11
FDD_TRK0_N 12
FDD_WP_N 13
FDD_RDATA_N 14
FDD_HEAD_N 15
FDD_CHANGE_N 18
MCU_MOTOR_B_N 15
MCU_DIR_N 14
MCU_STEP_N 13
MCU_WDATA_N 12
MCU_WGATE_N 11
U2
SN74LVC245AN / 5V→3V3 INPUT
3V3 1
GND 10
3V3 20
GND 19
FDD_HEAD_N 2
MCU_HEAD_N 3
FDD_DENSEL 4
MCU_DENSEL 17
DOCK_RX_RAW 5
DOCK_RX 18
GND 9
GND 6
GND 8
GND 7
GND 17
GND 15
GND 16
J2
FLOPPY POWER 5V GND GND 12V
VS_IN 1
GND 2
GND 3
Q4
TN0702 / FDD INDEX
GND 1
ASSERT_INDEX 2
FDD_INDEX_N 3
Q5
TN0702 / FDD TRK0
GND 1
ASSERT_TRK0 2
FDD_TRK0_N 3
Q6
TN0702 / FDD WP
GND 1
ASSERT_WP 2
FDD_WP_N 3
Q7
TN0702 / FDD RDATA
GND 1
ASSERT_RDATA 2
FDD_RDATA_N 3
Q8
TN0702 / FDD CHANGE
GND 1
ASSERT_CHANGE 2
FDD_CHANGE_N 3
RN1
8x10k BUSSED / COMMON PINS
VS 1
FDD_HEAD_N 2
FDD_DS_B_N 3
FDD_DS_A_N 4
FDD_MOTOR_B_N 5
FDD_DIR_N 8
FDD_STEP_N 7
FDD_WDATA_N 6
FDD_WGATE_N 5
RN2
8x10k BUSSED / COMMON PINS
VS 1
FDD_HEAD_N 2
FDD_DENSEL 3
NC 4
NC 5
NC 6
NC 7
NC 8
RN3
8x1k BUSSED / COMMON GND
GND 1
ASSERT_INDEX 2
ASSERT_TRK0 3
ASSERT_WP 4
ASSERT_RDATA 5
ASSERT_CHANGE 8
DOCK_RX_RAW 7
MEDIA_ARM 6
INTERLOCK_OK 5
F1
2A FAST / LOW DROP
VS_IN 1
VS 2
D1
1N5819 / USB BACKFEED BLOCK
VSYS 1
VS 2
C1
4.7uF 16V
VS 1
GND 2
C2
100nF 50V
3V3 1
GND 2
C3
100nF 50V
3V3 1
GND 2
C5
100nF 50V
3V3 1
GND 2
J3
1725708 / CONTACT WIRES 1 TO 7
GND 1
MEDIA_3V3 2
MEDIA_CS_N 3
MEDIA_SCK 4
MEDIA_MOSI 5
MEDIA_MISO 6
NC 7
J4
DOCK LINK 3X3 UART
GND 1
VS 2
DOCK_TX 3
DOCK_RX_RAW 4
DOCK_BUSY 5
LED_DATA 9
3V3 8
GND 7
NC 6
NC 10
R1
10k SPI CS DEFAULT
3V3 1
SPI_CS_N 2
R2
10k MEDIA ERROR PULLUP
3V3 1
MEDIA_ERROR_N 2
J5
3X3 OUT / INTERLOCK RETURN
3V3 1
INTERLOCK_OK 2
U4
SN74LVC245AN / SWITCHED MEDIA ISOLATION
MEDIA_3V3 1
SPI_CS_N 2
SPI_SCK 3
SPI_MOSI 4
MEDIA_MISO 5
GND 6
GND 7
GND 8
GND 9
GND 10
NC 11
NC 12
NC 13
NC 14
SPI_MISO 15
MEDIA_MOSI 16
MEDIA_SCK 17
MEDIA_CS_N 18
MEDIA_OE_N 19
MEDIA_3V3 20
U6
LP2951ACN / SWITCHED 3V1 100mA
MEDIA_3V3 1
NC 2
MEDIA_OFF 3
GND 4
MEDIA_ERROR_N 8
NC 7
MEDIA_FB 6
VS_MEDIA 5
R4
10k 1%
MEDIA_3V3 1
MEDIA_FB 2
R5
10k 1%
MEDIA_FB 1
GND 2
R6
68R FAILSAFE BLEED
MEDIA_3V3 1
MEDIA_3V3 2
R7
680R DEFAULT OFF
VS 1
MEDIA_OFF 2
R8
2R2 DISCHARGE
MEDIA_3V3 1
DISCHARGE_C 2
R9
10k MSO DEFAULT
SPI_MISO 1
GND 2
R10
1 OHM CAP ESR
MEDIA_CAP_RETURN 1
GND 2
R11
2k2 ARM GATE
MEDIA_ARM 1
ARM_BASE 2
R12
2k2 INTERLOCK GATE
INTERLOCK_OK 1
LOCK_BASE 2
Q1
IRLZ44N / HARD DISCHARGE
MEDIA_OFF 4
DISCHARGE_C 2
GND 3
Q2
TN0702 / ARM PERMISSION
PERMISSION_MID 1
ARM_BASE 2
MEDIA_OFF 3
Q3
TN0702 / INTERLOCK PERMISSION
GND 1
LOCK_BASE 2
PERMISSION_MID 3
C6
22uF 16V / OUTPUT
MEDIA_3V3 1
MEDIA_CAP_RETURN 2
C7
100nF 50V
MEDIA_3V3 1
GND 2
C9
1uF 50V X7R
VS 1
GND 2
Q9
SUP53P06 / MEDIA INPUT CUT
MEDIA_OFF 1
VS_MEDIA 2
VS 3
Q10
TN0702 / OE ARM
OE_MID 1
ARM_BASE 2
MEDIA_OE_N 3
Q11
TN0702 / OE INTERLOCK
GND 1
LOCK_BASE 2
OE_MID 3
R13
10k OE DEFAULT OFF
MEDIA_3V3 1
MEDIA_OE_N 2
R1: Standard IBM-PC 34-pin FDD rear interface; cable twist handled by firmware selection.
5V inputs use 5.5V-tolerant LVC245 at 3.3V; Outputs are TN0702 open drain: ASSERT=1 pulls cable LOW.
J3 is the isolated seven-wire contact output. J5 series seal-NO/eject-NC loop MUST open before disk motion.
Pico is a factory-assembled SMD module plugged into TH sockets. Carrier assembly is through-hole.
Legacy floppy encoding/track streaming and native dock protocol require new firmware; no compatibility claim.
PROTOTYPE / NOT RELEASED / electrical and mechanical bench tests required
Sheet: /
File: cyber-drive-controller-r1.kicad_sch
Title: RODDY CYBER DRIVE / CORE CONTROLLER
Size: A1
Date:
Rev: R1.1 hardware finish
KiCad E.D.A. 10.0.6
Id: 1/1